Business case

What Kept is worth to Maple Bank, from the bank's own data

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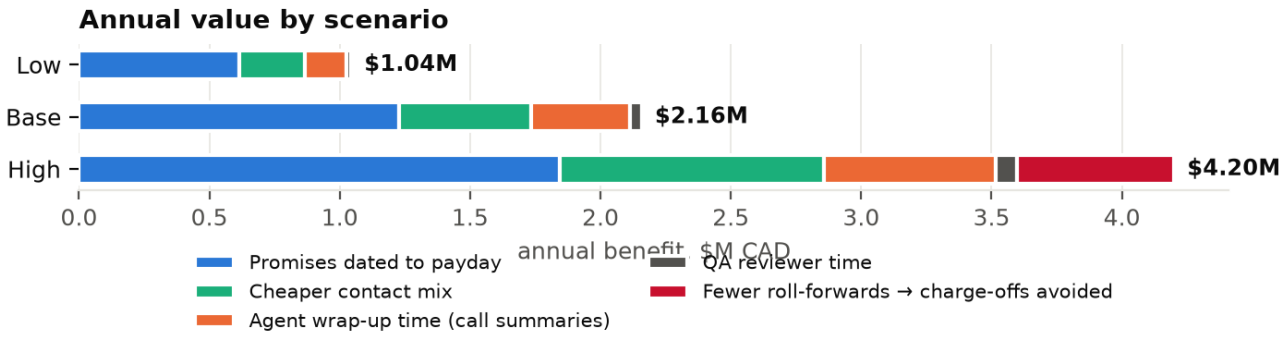
TL;DR

Base case: $2.2M a year (low $1.0M, high $4.2M) against **$150K a year to run** and **$300K to productionise**. First-year ROI **3.8x** in the base case (1.3x low), payback in about **2 months** (4 months low).

Where it comes from, biggest first:

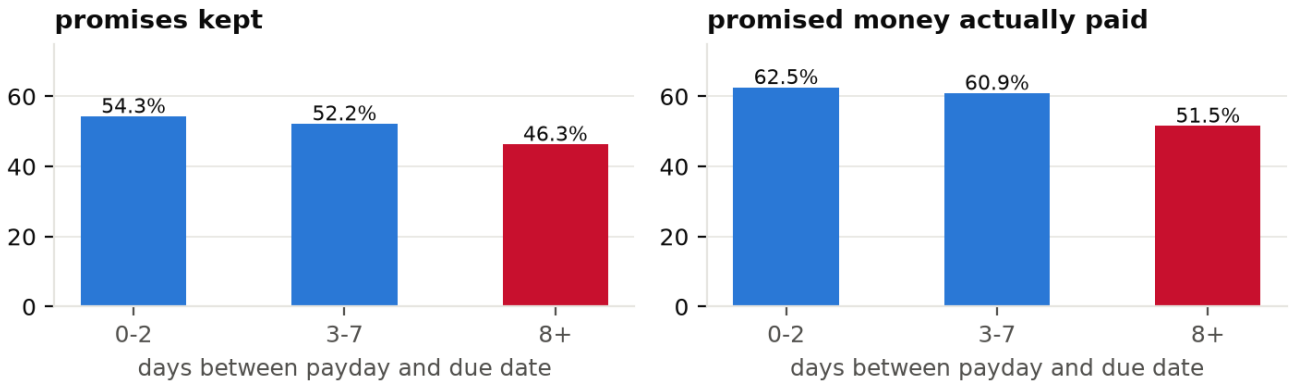
1. **Promises dated to payday.** Promises due 0–2 days after payday get **62.5%** of the promised money paid; 8+ days after, **51.5%**. Kept sizing puts the due date right after payday.
2. **A cheaper, kinder contact mix.** Same number of calls, but most become hardship reviews instead of collection calls, and $1.10 letters become near-free texts, emails and app pushes.
3. **Agent time.** Auto call summaries cut after-call work on 629,435 agent calls a year.
4. **QA time.** Every call gets pre-scored, so reviewers listen to the ones that need it.

Plus an upside we don't count in the base case: **every point of roll rate is worth ~$603K a year** in charge-offs avoided. The 5% hold-out measures it after launch.



1. The payday effect (the core idea)

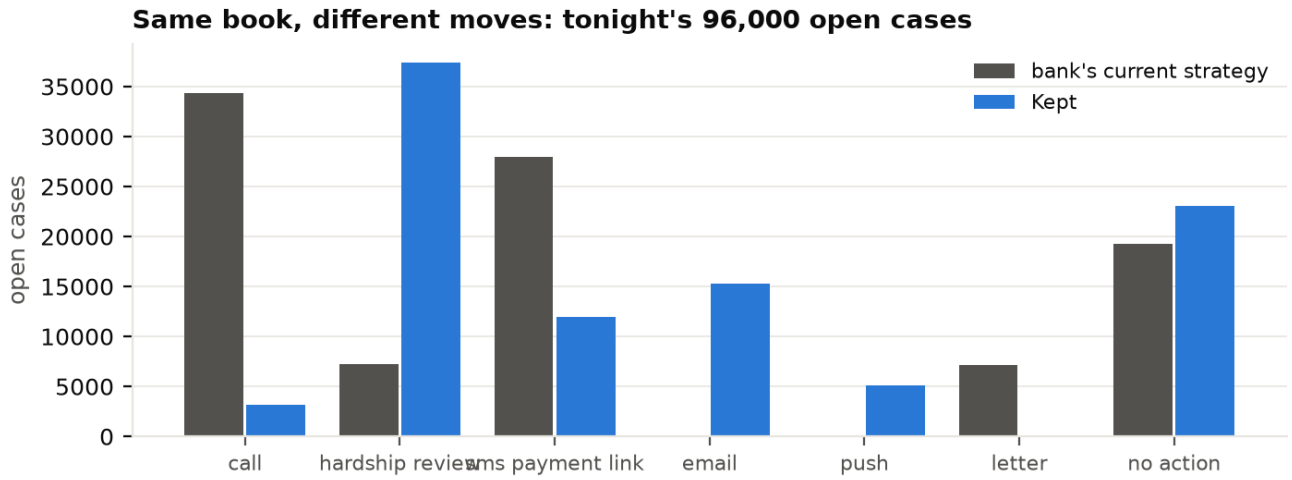
From 31,957 resolved promises where we can see the customer's salary deposits:



Promise due	Promises	Kept	Promised money paid
0-2 days after payday	8,359	54.3%	62.5%
3-7 days after payday	7,553	52.2%	60.9%
8+ days after payday	16,045	46.3%	51.5%

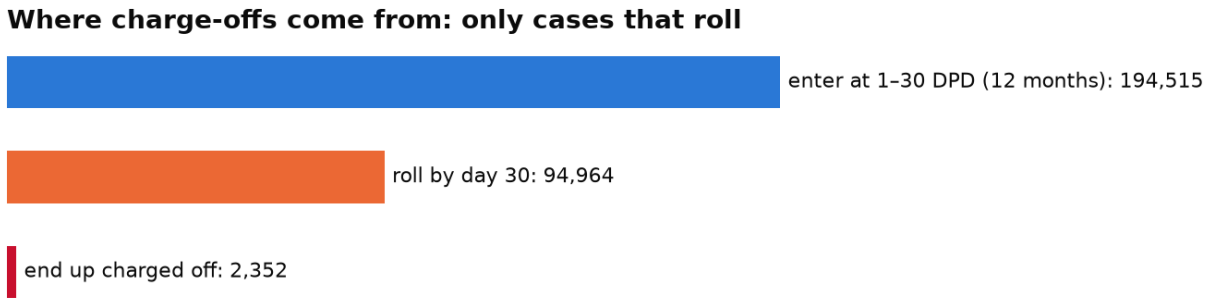
- **The gap:** weighting by where the money sits today, moving promises that fall 3+ days after payday to 0–2 days lifts the paid share by **5.9 points**.
- **The base:** **$41.7M** promised in the 12 months to 28 Sep 2026.
- **We only take half of it** in the base case (`PAYDAY_CAPTURE` = 50%): the gap is descriptive, and some of it is who gets which due date, not the date itself. That gives **$1.23M a year**; the full gap would be $2.46M.

2. Contact mix and cost



- **Calls:** 40,601 call-type actions for Kept vs 41,574 for the current strategy. Basically the same number, but **37,409 of ours are hardship reviews** (a specialist conversation) where the old strategy has 34,326 collection calls.
- **Letters → digital:** the old strategy sends 7,180 letters at $1.10; Kept uses SMS links ($0.01), email and app pushes ($0.001).
- **Cost per pass over the book** (channel costs from `contact_history`, Sep 2026): $80,927 old vs $71,187 Kept, **$9,740 saved each pass**. At 52 passes a year (base) that's **$506K**.
- **Model's view:** tonight's 55,674 positive-uplift proposals are expected to bring in **$2.06M more within 14 days than waiting** (uplift × balance at risk). It's a model estimate, so it's not in the totals: the hold-out proves or disproves it.

3. Roll rate → charge-offs (the upside)



- **48.8%** of cases that enter at 1–30 days past due have rolled by day 30 (330,779 cases since 2016).
- Of rolled cases, **2.48%** end up charged off, averaging **$12,523**. Of cases that don't roll: **0.00%**. Rolling is the only road to a charge-off.
- **194,515 cases ($63.9M overdue)** entered at 1–30 DPD in the last 12 months, so **1 point of roll rate ≈ $603K a year** avoided.
- The design's hypothesis is 1–2 points. We keep it **out of the base case** because it overlaps the payday gain (kept promises are what stop rolls), and count 1 point in the high case only.

4. Time saved

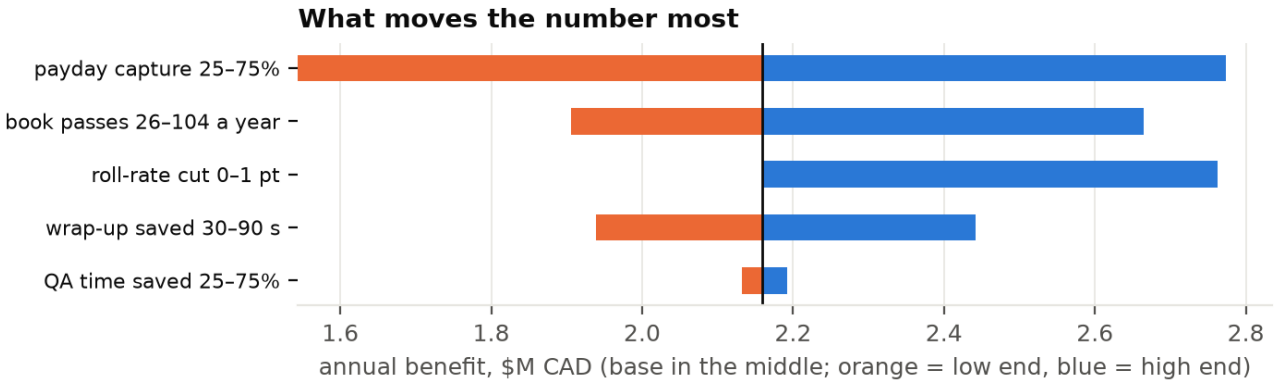
- **Call summaries:** 629,435 agent calls in 12 months, average handle time 6.8 min. Saving 60 s of after-call work each at $36/h is **$378K a year** (about 5.8 agents' worth of hours back on the phones).

- **QA:** a 2% manual sample at 12 min a call is 2,518 reviewer hours a year. Pre-scoring every call (88–100% agreement with human QA per item) lets reviewers skip the clean ones: **$45K** at 50% time saved, and coverage goes from 2% of calls to all of them.

5. Cost, ROI, payback

	Low	Base	High
Promises dated to payday	$614,603	**$1,229,207**	$1,843,810
Cheaper contact mix	$253,239	**$506,479**	$1,012,957
Agent wrap-up time (call summaries)	$157,359	**$377,661**	$660,907
QA reviewer time	$18,883	**$45,319**	$79,309
Fewer roll-forwards → charge-offs avoided	$0	**$0**	$603,362
Total benefit / year	**$1,044,084**	**$2,158,666**	**$4,200,345**
Run cost / year	$150,000	$150,000	$150,000
Build (one-off)	$300,000	$300,000	$300,000
Year-1 ROI	**1.3x**	**3.8x**	**8.3x**
Payback	4.0 months	1.8 months	0.9 months

Why it's cheap to run. One nightly batch (~7 minutes for the whole book on one machine), a local open model for text (no per-call fees, notes never leave the bank), a hosted model only for the Ask, and DuckDB/dbt instead of a cluster. The run cost is mostly one ops person.



Assumptions (all in `scripts/business_case.py`, change them and re-run):

Assumption	Base	Low	High	Why
Payday gap we capture	50%	25%	75%	the gap is descriptive; half is a cautious read
Passes over the open book a year	52	26	104	weekly work-through under the 3-in-7-days cap
After-call work saved per call	60 s	30 s	90 s	the summary pre-fills the wrap-up note
Loaded agent cost	$36/h	$30/h	$42/h	~$50K salary + 30% on-cost
QA reviewer time saved	50%	25%	75%	reviewers skip pre-scored clean calls
Roll-rate points counted	0	0	1	overlaps payday; high case only
Build / run cost	$300K / $150K			4 engineers × 3 months; 1 ops FTE + infra

6. Things that don't fit in a spreadsheet

- **Fair treatment and regulator risk.** Hardship shows up when the customer *says* it (text) or the money *shows* it (late payroll, a pay cut, new EI), and it immediately limits contact to protective actions, with two people releasing anything sensitive. Every decision has a plain-English reason and an audit trail.
- **The cost of the protection, honestly.** 38,115 of tonight's cards are Red (a specialist reviews first). That's real specialist time. A rule for customers already on an active hardship plan would cut it; it's the next thing we'd ship.
- **Measured, not promised.** A permanent 5% hold-out stays on the old strategy, so every number above gets checked against reality, and a kill switch reverts everyone if it isn't working.

Rebuild: `uv run python scripts/business_case.py` (reads only the raw files and the cards export).